WORKED SOLUTION

STEP II Specimen paper, Question 3

Pure • De Moivre's Theorem

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q3

The key idea

De Moivre's theorem is a natural induction: multiplying by one more factor of $\cos\theta + \mathrm{i}\sin\theta$ adds θ to the angle, by the compound-angle formulae. For the equations, the pattern $\cos ky$, $\sin ky$ next to x^k shouts "real and imaginary parts". Treat the first equation as the real part and the second as the imaginary part of a single complex equation in $z = x(\cos y + \mathrm{i}\sin y)$, and the pair collapses to a cubic in z with an easy root.

Solution

De Moivre's theorem

We use induction on n. For $n = 1$ both sides are $\cos\theta + \mathrm{i}\sin\theta$, so the result holds.

Suppose it holds for some $n = k \geq 1$. Then

$$\begin{aligned}(\cos\theta + \mathrm{i}\sin\theta)^{k+1} &= (\cos k\theta + \mathrm{i}\sin k\theta)(\cos\theta + \mathrm{i}\sin\theta)\\ &= (\cos k\theta\cos\theta - \sin k\theta\sin\theta) + \mathrm{i}(\sin k\theta\cos\theta + \cos k\theta\sin\theta)\\ &= \cos(k+1)\theta + \mathrm{i}\sin(k+1)\theta,\end{aligned}$$

using the induction hypothesis in the first line, $\mathrm{i}^2 = -1$ in the second, and the compound-angle formulae for $\cos(A+B)$ and $\sin(A+B)$ in the third. So the result holds for $n = k+1$, and by induction it holds for every positive integer n.

Solving the equations

Let x and y be real and put

$$z = x(\cos y + \mathrm{i}\sin y).$$

By de Moivre's theorem, for $n = 1, 2, 3$,

$$z^n = x^n(\cos y + \mathrm{i}\sin y)^n = x^n\cos ny + \mathrm{i}\,x^n\sin ny.$$

(This holds whatever the sign of x: the real factor x simply comes out as x^n.) Since x and y are real, the left-hand sides of the two equations are exactly the real and imaginary parts of z^3+2z^2+2z. So the pair of real equations is equivalent to the single complex equation

$$z^3+2z^2+2z=-1, \qquad \text{that is,} \qquad z^3+2z^2+2z+1=0.$$

Ask yourself Why is one complex equation the same as the two real ones?

Two complex numbers are equal exactly when their real parts agree and their imaginary parts agree. The right-hand side -1 has real part -1 and imaginary part 0, which are the two right-hand sides we were given. So (x,y) satisfies both real equations if and only if z satisfies the cubic.

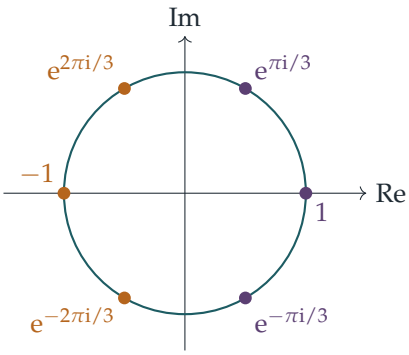
Now $z=-1$ is a root, since $-1+2-2+1=0$, and dividing out gives

$$z^3+2z^2+2z+1=(z+1)(z^2+z+1).$$

The quadratic has roots $z=\frac{1}{2}(-1\pm\mathrm{i}\sqrt{3})$. So the three roots are

$$z=-1=\cos\pi+\mathrm{i}\sin\pi, \qquad z=\cos\tfrac{2\pi}{3}\pm\mathrm{i}\sin\tfrac{2\pi}{3},$$

all on the unit circle.



It remains to turn each root back into real pairs (x,y). Since $|z|=|x|$, every solution has $|x|=1$; in particular $x=0$ is impossible (it would give $0=-1$).

- If $x=1$, then $\cos y+\mathrm{i}\sin y=z$, so y is an argument of one of the three roots: $y=\pi$, $\frac{2\pi}{3}$ or $-\frac{2\pi}{3}$, plus any multiple of 2π.
- If $x=-1$, then $\cos y+\mathrm{i}\sin y=-z$, which is one of 1, $\cos\frac{\pi}{3}\mp\mathrm{i}\sin\frac{\pi}{3}$ (the plum points in the figure). So $y=0$, $-\frac{\pi}{3}$ or $\frac{\pi}{3}$, plus any multiple of 2π.

Each of these gives a root of the cubic, so all of them work. With k any integer, the solutions are

$$x=1,\quad y=\pi+2k\pi \text{ or } y=\pm\tfrac{2\pi}{3}+2k\pi; \qquad x=-1,\quad y=2k\pi \text{ or } y=\pm\tfrac{\pi}{3}+2k\pi.$$

In $[0,2\pi)$ that is six pairs: $(1,\frac{2\pi}{3})$, $(1,\pi)$, $(1,\frac{4\pi}{3})$, $(-1,0)$, $(-1,\frac{\pi}{3})$, $(-1,\frac{5\pi}{3})$.

Checks

Try $x = -1$, $y = 0$: the first equation gives $-1 + 2 - 2 = -1$ and the second is $0 = 0$. Try $x = 1$, $y = \frac{2\pi}{3}$: the cosines are $1, -\frac{1}{2}, -\frac{1}{2}$, giving $1 - 1 - 1 = -1$; the sines are $0, -\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}$, giving $0 - \sqrt{3} + \sqrt{3} = 0$. A computer search over a fine grid of $x \in [-3, 3]$, $y \in [0, 2\pi)$ finds near-solutions only at the six points above (to rounding), so none has been missed.

Common slips

- Assuming x is a modulus and so $x \geq 0$. The question allows any real x, and the $x = -1$ solutions are just as valid; they are the same points z described with y shifted by π.
- Forgetting the $+2k\pi$: y is only determined modulo 2π, so there are infinitely many pairs (x, y).

Beyond STEP

Packaging two real equations as one complex one is a standard move: the real and imaginary parts of a polynomial in $z = re^{i\theta}$ satisfy exactly this kind of trigonometric system, and the factorisation $z^3 + 2z^2 + 2z + 1 = (z + 1)(z^2 + z + 1)$ places the roots on the unit circle. In a first course you will meet $e^{i\theta}$ defined by a power series, which makes de Moivre's theorem hold for every integer (and gives a meaning to non-integer powers, with care over branches). Needham's *Visual Complex Analysis* is a beautiful geometric account of all this.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.